

Lecture#03

Odds

Odds express how likely an event is **in comparison to it not happening**. Odds focus on the **ratio**. It is used for comparison.

$$\text{Odds}(A) = \frac{P(\text{Event } A)}{P(\text{Event } \bar{A})} = \frac{P(\text{Event } A)}{1 - P(\text{Event } A)}$$

[Basically for two outcomes, where $P(A) + P(\bar{A}) = 1$]

So odds compare *success vs failure*.

Types of odds

a) Odds in favor of an event A

$$\text{Odds in favor A} = \frac{\text{number of favorable outcomes}}{\text{number of unfavorable outcomes}} = \frac{\frac{n(A)}{n(S)}}{\frac{n(\bar{A})}{n(S)}} = \frac{n(A)}{n(\bar{A})}$$

b) Odds against an event A

$$\text{Odds against A} = \frac{\text{number of unfavorable outcomes}}{\text{number of favorable outcomes}} = \frac{\frac{n(\bar{A})}{n(S)}}{\frac{n(A)}{n(S)}} = \frac{n(\bar{A})}{n(A)}$$

Where,

$n(A)$ = number of favorable outcomes,

$n(\bar{A}) = n(B)$ = number of unfavorable outcomes, and

$n(S)$ = total sample samce = $n(A) + n(\bar{A})$

Odds in favor and odds against are reciprocals.

Suppose A and $B = \bar{A}$ are two events. If $P(A) = p$, then Odds in favor of A, $O_A = \frac{p}{1-p}$.

Again, it can be also said that Odds against B, $O_A = \frac{p}{1-p}$.

Similarly, Odds in favor of B, $O_B = \frac{1-p}{p}$. Again, it can be also said that Odds against A, $O_{\bar{A}} = \frac{1-p}{p}$.

If odds in favor of A are **a: b**, then probabilities are, $P(A) = \frac{a}{a+b}$ and $P(\bar{A}) = \frac{b}{a+b}$.

Example 1. A bag has some red balls and some blue balls.

Odds in favor of red are 3: 7. Find the probabilities of red and blue balls.

Solution:

Here, $n(A)$ = number of red balls = 3, and $n(\bar{A})$ = number of blue balls = 7. So,
 $n(S) = n(A) + n(\bar{A}) = 3 + 7 = 10$

Therefore, probability of red: $P(A) = \frac{3}{10} = 0.3$ and probability of blue: $P(\bar{A}) = \frac{7}{10} = 0.7$

Example 2. The odds in favor of an event A are 5:3. Find the probability of A.

Solution

Probability of A: $P(A) = \frac{5}{8}$.

Example 3. The probability of an athlete winning a race is 0.2. Find the odds against winning.

Solution

Given, Probability of winning, $P(W) = 0.2$. So, probability of not winning is, $P(\bar{W}) = 0.8$.

Now, Odds against W, $O_{\bar{W}} = \frac{P(\bar{W})}{P(W)} = \frac{0.8}{0.2} = \frac{4}{1} = 4$.

So the odds against winning is 4 or you can write 4:1. For interpretation, you can say... The odds against winning are 4 to 1, which means that for every 1 chance the athlete wins, there are 4 chances that the athlete does not win.

Problem 1. An engineering statistics problem is given to two students, X and Y. The odds against student X solving the problem are 8:6, and the odds in favor of student Y solving the problem are 14:16. (Assume X and Y work independently).

(a) Find the probability that both X and Y solve the problem.

(b) Find the probability that neither solves the problem.

- (c) Find the probability that at least one solves the problem.
- (d) Find the probability that either X or Y solves the problem.
- (e) Find the probability that exactly one solves the problem.

Solution:

Let, **X solves** = X , and **X fail to solve** = $\bar{X}$. Similarly, let, **Y solves** = Y , and **Y fail to solve** = $\bar{Y}$.

Given, Odds **against** X solving the problem = 8 : 6. So, $P(\bar{X}) = \frac{8}{8+6} = \frac{4}{7}$, $P(X) = \frac{6}{8+6} = \frac{3}{7}$

Again, odds **in favor of** Y solving the problem = 14 : 16. So, $P(Y) = \frac{14}{14+16} = \frac{7}{15}$,

$$P(\bar{Y}) = \frac{16}{14+16} = \frac{8}{15}.$$

(i) Probability that they both solve the problem

As X and Y work independently, so we know for independence events,

$$P(X \cap Y) = P(X) \times P(Y)$$

Therefore, $P(\text{both solve}) = P(X \cap Y) = P(X \text{ solves}) \times P(Y \text{ solves}) = P(X) \times P(Y)$

$$= \left(\frac{3}{7}\right) \times \left(\frac{7}{15}\right) = \frac{3}{15} = \frac{1}{5}$$

(ii) Probability that neither solves the problem

$$\begin{aligned} P(\text{neither solves}) &= P(\bar{X} \cap \bar{Y}) = P(X \text{ fails}) \times P(Y \text{ fails}) = P(\bar{X}) \times P(\bar{Y}) \\ &= \left(\frac{4}{7}\right) \times \left(\frac{8}{15}\right) = \frac{32}{105} \end{aligned}$$

Or, $P(\text{neither solves}) = 1 - P(X \cup Y)$

(iii) Probability that at least one solves

At least one solve means X solves or Y solves or both solve, i.e., we need to find $P(X \cup Y)$.

We know that, $P(X \cup Y) = P(X) + P(Y) - P(X \cap Y)$

or $P(X \cup Y) = 1 - P(\overline{X \cup Y}) = 1 - P(\bar{X} \cap \bar{Y})$ [Using De Morgan's Law].

Therefore, $P(X \cup Y) = 1 - P(\bar{X} \cap \bar{Y}) = 1 - \frac{32}{105} = \frac{73}{105}$.

(iv) Probability that either X or Y solves

Similar to (iii) because it means either X solves, Y solves, or both solve

(v) Probability that exactly one solves the problem

$P(\text{exactly one})$

$$\begin{aligned} &= P(X \text{ solves and } Y \text{ does not solve}) \\ &+ P(X \text{ does not solve and } Y \text{ solves}) = P(X \cap \bar{Y}) + P(\bar{X} \cap Y) \\ &= P(X)P(\bar{Y}) + P(\bar{X})P(Y) = \frac{3}{7} \times \frac{8}{15} + \frac{4}{7} \times \frac{7}{15} = \frac{52}{105}. \end{aligned}$$

Or,

$$P(\text{exactly one}) = P(X \cup Y) - P(X \cap Y) = \frac{73}{105} - \frac{1}{5} = \frac{73}{105} - \frac{21}{105} = \frac{52}{105}$$

Problem 2 (Similar problem, but dependent case)

The odds against student X solving a business statistics problem are 8:6. The odds in favor of student Y solving the problem are 14:16. In addition, suppose that if X solves the problem, then the probability that Y also solves it is 60%.

(a) Find the probability that both X and Y solve the problem.

(b) Find the probability that neither solves the problem.

(c) Find the probability that at least one solves the problem.

(d) Find the probability that either X or Y solves the problem.

(e) Find the probability that exactly one solves the problem.

Solution:

Let, **X solves** = X , and **X fail to solve** = $\bar{X}$. Similarly, let, **Y solves** = Y , and **Y fail to solve** = $\bar{Y}$.

Given, Odds **against** X solving the problem = 8 : 6. So, $P(\bar{X}) = \frac{8}{8+6} = \frac{4}{7}$, $P(X) = \frac{6}{8+6} = \frac{3}{7}$

Again, odds **in favor of Y** solving the problem = 14 : 16. So, $P(Y) = \frac{14}{14+16} = \frac{7}{15}$,

$$P(\bar{Y}) = \frac{16}{14+16} = \frac{8}{15}.$$

$$\text{and } P(Y|X) = 0.6 = \frac{3}{5}$$

(i) Probability that they both solve the problem

As X and Y do not work independently, so we know for dependent events,

$$P(X \cap Y) = P(X) \times P(Y|X)$$

Therefore, $P(\text{both solve}) = P(X \cap Y) = P(X) \times P(Y|X)$

$$= \left(\frac{3}{7}\right) \times \left(\frac{3}{5}\right) = \frac{9}{35}$$

(ii) Probability that neither solves the problem

$$P(\text{neither solves}) = P(\bar{X} \cap \bar{Y}) = P(\overline{X \cup Y}) = 1 - P(X \cup Y)$$

$$= 1 - \{P(X) + P(Y) - P(X \cap Y)\} = 1 - \frac{67}{105} = \frac{38}{105}.$$

(iii) Probability that at least one solves

At least one solve means X solves or Y solves or both solve, i.e., we need to find $P(X \cup Y)$.

$$\text{We know that, } P(X \cup Y) = P(X) + P(Y) - P(X \cap Y) = \frac{67}{105}$$

(iv) Probability that either X or Y solves

Similar to (iii) because it means either X solves, Y solves, or both solve

(v) Probability that exactly one solves the problem

$$P(\text{exactly one}) = P(X \cup Y) - P(X \cap Y) = \frac{67}{105} - \frac{9}{35} = \frac{40}{105}.$$

Note. $Y = (Y \cap X) \cup (Y \cap \bar{X}) \Rightarrow P(Y) = P(Y \cap X) + P(Y \cap \bar{X})$, as $(Y \cap X)$ and $(Y \cap \bar{X})$ are mutually exclusive or disjoint.

Joint Probability

Joint probability deals with the probability of two or more events happening simultaneously.

For two events A and B, the joint probability is written as: **$P(A \cap B)$** . This represents the probability that both A and B occur at the same time.

Joint Probability for Independent Events

If A and B are independent: $P(A \cap B) = P(A) \times P(B)$

Example: Toss a coin and roll a die.

Event A: coin shows heads ($P(A) = \frac{1}{2}$)

Event B: die shows 4 ($P(B) = \frac{1}{6}$)

Joint probability: $P(A \cap B) = \frac{1}{2} \times \frac{1}{6} = \frac{1}{12}$

Joint Probability for Dependent Events

If A and B are dependent: **$P(A \cap B) = P(A) \times P(B | A)$** if A happen first or

$P(A \cap B) = P(B) \times P(A | B)$ if B happen first.

Example: Drawing two cards without replacement from a deck:

Event A: first card is an Ace ($P(A) = \frac{4}{52}$)

Event B: second card is an Ace ($P(B|A) = \frac{3}{51}$)

Joint probability: $P(A \cap B) = P(A) \times P(B | A) = \frac{4}{52} \times \frac{3}{51} = \frac{1}{221}$

Problem 1

In a class, 60% of students study **Mathematics**, 50% study **Physics**, and 30% study **both**. Find:

1. Probability that a student studies Mathematics or Physics.
2. Probability that a student studies Mathematics but not Physics.
3. Probability that a student studies neither Mathematics nor Physics.

Solution

Given: $P(M) = 0.6, P(P) = 0.5, P(M \cap P) = 0.3$

1. $P(\mathbf{M \text{ or } P}) = P(M) + P(P) - P(M \cap P) = 0.6 + 0.5 - 0.3 = 0.8$
2. $P(\mathbf{M \text{ and not } P}) = P(M) - P(M \cap P) = 0.6 - 0.3 = 0.3$
3. $P(\mathbf{\text{neither } M \text{ nor } P}) = 1 - P(M \cup P) = 1 - 0.8 = 0.2$

Problem 2

A bag contains 3 red, 4 green, and 5 blue balls. Two balls are drawn **without replacement**. Find:

1. Probability that both are red.
2. Probability that one red and one green.
3. Probability that at least one blue.

Solution

Given, red=3 ball, green=4 balls, and blue=5 balls. So, total balls = 12

Total pairs = $12C_2 = 66$

1. $P(\mathbf{\text{Both red}}) = \frac{{}^3C_2}{{}^{12}C_2} = \frac{3}{66} = \frac{1}{22}$.
2. $P(\mathbf{\text{One red and one green}}) = \frac{{}^3C_1 \times {}^4C_1}{{}^{12}C_2} = \frac{12}{66} = \frac{2}{11}$.
3. $P(\mathbf{\text{At least one blue}}) = 1 - P(\text{no blue}) = 1 - \frac{{}^7C_2}{{}^{12}C_2} = 1 - \frac{21}{66} = 1 - \frac{7}{22} = \frac{15}{22}$.

Lecture #04

Joint Probability Table

For **discrete** events, a **joint probability table** (2×2) can help visualize probabilities of all combinations.

	Y=Yes	$\bar{Y}$ =No	Total				
X=Yes	$n(X \cap Y)$	$n(X \cap \bar{Y})$	$n(X)$	=>	X=Yes	$P(X \cap Y)$	$P(X \cap \bar{Y})$
$\bar{X}$ =No	$n(\bar{X} \cap Y)$	$n(\bar{X} \cap \bar{Y})$	$n(\bar{X})$		$\bar{X}$ =No	$P(\bar{X} \cap Y)$	$P(\bar{X} \cap \bar{Y})$
Total	$n(Y)$	$n(\bar{Y})$	$n(S)$		Total	$P(Y)$	$P(\bar{Y})$
							1

Each cell = joint probability of row and column event.

Discrete and Continuous Variable Examples....

Problem 01: In an office of 100 employees, 75 read English, 50 read Bangla dailies and 40 read both. An employee is selected at random. What is the probability that the selected employee

(a) reads English newspaper? (b) reads at least one of the papers? (c) reads none? (d) reads Bangla but not English?

Solution: Let us define the above events,

E = Reads English, B = Reads Bangla, $\bar{E} \cap \bar{B}$ = Reads none,

$B \cap \bar{E}$ = Reads Bangla but not English.

The number of the cases favorable to the above events can be placed in a tabular form as follows:

	E	$\bar{E}$	Total
B	$n(B \cap E) = 40$	$n(B \cap \bar{E}) = ?$	$n(B) = 50$
$\bar{B}$	$n(\bar{B} \cap E) = ?$	$n(\bar{B} \cap \bar{E}) = ?$	$n(\bar{B}) = 50$
Total	$n(E) = 75$	$n(\bar{E}) = 25$	$n(S) = 100$

(a) The probability that selected employee reads English is,

$$P(E) = \frac{n(E)}{n(S)} = \frac{75}{100} = 0.75$$

(b) The probability that selected employee reads at least one is:

$$\begin{aligned} P(E \cup B) &= P(E) + P(B) - P(E \cap B) \\ &= \left(\frac{75}{100} + \frac{50}{100} - \frac{40}{100} \right) = \frac{85}{100} = 0.85 \end{aligned}$$

(c) The probability that selected employee reads none is:

$$P(\bar{B} \cap \bar{E}) = P(\overline{(E \cup B)}) = 1 - P(E \cup B) = (1 - 0.85) = 0.15$$

(d) The probability that selected employee reads Bangla but not English is:

$$P(B \cap \bar{E}) = \frac{n(B) - n(B \cap E)}{n(S)} = \frac{50 - 40}{100} = \frac{10}{100} = 0.10$$

Problem 02: Of the total students at a women's college, 60% wear neither a ring nor a necklace, 20% wear a ring, and 30% wear a necklace. If one of the women is randomly chosen, find the probability that she is wearing (a) a ring or a necklace (b) Both.

Solution:

Let R and N respectively denote the events that a woman wears a ring and a necklace. We are given that, $P(R) = 0.20$, $P(N) = 0.30$, $P(\bar{R} \cap \bar{N}) = 0.60$

(a) The probability that she is wearing a ring or a necklace is:

$$P(R \cup N) = 1 - P(\overline{R \cup N}) = 1 - P(\bar{R} \cap \bar{N}) = 1 - 0.60 = 0.40$$

(b) The probability that she wears both is:

$$P(R \cap N) = P(R) + P(N) - P(R \cup N) = 0.20 + 0.30 - 0.40 = 0.10$$

Problem 03: A class contains 10 men and 20 women of whom half of the men and half of the women have brown eyes. A person is chosen at random. What is the probability that the person is either a man or has brown eyes?

Solution: To solve this problem, we can construct a table as follows:

	Man (M)	Woman (W)	Total
Brown (B)	$n(M \cap B) = 5$	$n(W \cap B) = 10$	$n(B) = 15$
Not brown ($\bar{B}$)	$n(M \cap \bar{B}) = 5$	$n(W \cap \bar{B}) = 10$	$n(\bar{B}) = 15$
Total	$n(M) = 10$	$n(W) = 20$	$n(S) = 30$

We must compute $P(M \cup B)$ were,

$$P(M \cup B) = P(M) + P(B) - P(M \cap B)$$

From the above table we have,

$$P(M) = \frac{n(M)}{n(S)} = \frac{10}{30} = \frac{1}{3}$$

$$P(B) = \frac{n(B)}{n(S)} = \frac{15}{30} = \frac{1}{2}$$

$$P(M \cap B) = \frac{n(M \cap B)}{n(S)} = \frac{5}{30} = \frac{1}{6}$$

Thus,

$$P(M \cup B) = \left(\frac{1}{3} + \frac{1}{2} - \frac{1}{6} \right) = \frac{2}{3}$$

The probability that the person is either a man or has brown eyes is $\frac{2}{3}$.

Problem 4. A company has two machines, A and B, producing items. Machine A produces 40% of the items, Machine B produces 60%. 5% of items from Machine A are defective, and 10% of items from Machine B are defective.

- Construct the joint probability table.
- Find the probability that a randomly selected item is defective.
- Find the probability that the item is from Machine A given that it is defective.
- Find the probability that the item is non-defective and from Machine B.

Solution

Let, X = item is from Machine A ($X = 1$), $\bar{X}$ = item is from Machine B ($X=0$), Y = item is defective ($Y = 1$), and $\bar{Y}$ = item is not defective ($Y = 0$).

Given, $P(A) = 0.4$, so $P(B) = 0.6$; $P(Y | A) = 0.05$, so $P(\bar{Y} | A) = 0.95$; $P(Y | B) = 0.10 \rightarrow P(\bar{Y} | B) = 0.90$

1. Joint Probability Table

Use the definition: $P(X = i, Y = j) = P(X = i) \cdot P(Y = j | X = i)$

	Y	$\bar{Y}$	Total
X	$0.4 \times 0.05 = 0.02$	$0.4 \times 0.95 = 0.38$	0.4
$\bar{X}$	$0.6 \times 0.10 = 0.06$	$0.6 \times 0.90 = 0.54$	0.6
Total	0.08	0.92	1

2. Probability that an item is defective

$$P(Y = 1) = P(X \cap Y) + P(\bar{X} \cap Y) = 0.02 + 0.06 = 0.08$$

3. Probability that the item is from Machine A given defective

$$P(X | Y) = \frac{P(X \cap Y)}{P(Y)} = \frac{0.02}{0.08} = 0.25$$

So, if an item is defective, there is a 25% chance it came from Machine A.

4. Probability that the item is non-defective and from Machine B

$$P(\bar{X} \cap \bar{Y}) = 0.6 \cdot 0.9 = 0.54$$

Problem 5. Two ballpoint pens are selected at random from a box that contains 3 blue pens, 2 red pens, and 3 green pens. If X is the number of blue pens selected and Y is the number of red pens selected, find

- (i) The joint probability function $f(x, y)$.
- (ii) $P[(X, Y) \in A]$, where A is the region $\{(x, y) \mid x + y \leq 1\}$.
- (iii) Find the marginal distribution of X alone and of Y alone.

Solution

Drawing 2 pens without replacement from a box containing: 3 blue, 2 red, and 3 green.
Total = 8 pens.

Let, X = number of blue pens selected= $\{0,1,2\}$, Y = number of red pens selected= $\{0,1,2\}$

Total number of possible pairs, $8_{C_2} = 28$

(i). Joint probability distribution $f(x, y)$

$X = 0, Y = 0$ indicates no blue, no red $\rightarrow$ both are green.

$$P(X = 0, Y = 0) = \frac{{}^3C_0 \times {}^2C_0 \times {}^3C_2}{8_{C_2}} = \frac{1 \times 1 \times 3}{28} = \frac{3}{28}$$

So, the joint probability distribution $f(x, y)$ table is as follows

			Y		Total
		0	1	2	
	0	$\frac{3}{28}$	$\frac{6}{28}$	$\frac{1}{28}$	$\frac{10}{28}$
X	1	$\frac{9}{28}$	$\frac{6}{28}$	-	$\frac{15}{28}$
	2	$\frac{3}{28}$	-	-	$\frac{3}{28}$
Total		$\frac{15}{28}$	$\frac{12}{28}$	$\frac{1}{28}$	1

(ii) As $\{(x, y) \mid x + y \leq 1\}$,

$$\begin{aligned} P((X, Y) \in A) &= P(X = 0, Y = 0) + P(X = 1, Y = 0) + P(X = 0, Y = 1) \\ &= \frac{3}{28} + \frac{9}{28} + \frac{6}{28} = \frac{18}{28} = \frac{9}{14}. \end{aligned}$$

(iii)

Marginal distribution of X	Marginal distribution of Y
$P(X = 0) = \frac{10}{28} = \frac{5}{14}$	$P(Y = 0) = \frac{15}{28}$
$P(X = 1) = \frac{10}{28} = \frac{5}{14}$	$P(Y = 1) = \frac{12}{28} = \frac{3}{7}$
$P(X = 2) = \frac{3}{28} = \frac{3}{28}$	$P(Y = 2) = \frac{1}{28}$